BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 149 · Algorithms: Examination Techniques

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How do IDE error diagnostics help?

Q2. Which type suits a count of pupils?
A. Integer
B. Real
C. Boolean
D. Character

Q3. “A character data type can hold an entire arbitrary-length sentence as one character.” Correct this. Explain why.

Q4. A bill is quantity × unit price + fee. “Calculations always run left to right.” Correct this. How does operator precedence affect the result?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Read algorithm details — Distinguish a stated loop boundary, index and test from what the algorithm is intended to do.

Trace faithfully — Record each update and branch, including false conditions and early stops.

Identify prerequisites — Explain when searching and sorting methods are valid.

Link method to marks — An exam explanation should specify precise comparisons, movement or changes to stored data.

New learning

Read algorithm details: Distinguish a stated loop boundary, index and test from what the algorithm is intended to do.

Trace faithfully: Record each update and branch, including false conditions and early stops.

Identify prerequisites: Explain when searching and sorting methods are valid.

Link method to marks: An exam explanation should specify precise comparisons, movement or changes to stored data.

Worked example

Trace linear search for 9 in [4,6,9,2] with zero-based indexes.

1. Compare index 0: 4 ≠ 9.

2. Compare index 1: 6 ≠ 9.

3. Compare index 2: 9 = 9; stop.

4. Found at index 2 after three comparisons.

We Do · Practise with guidance

Guided 1

Trace binary search for 14 in sorted [2,6,10,14,18,22,26].

Worked method: Index 3 found with one comparison.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the index to print the Program Counter abbreviation.

```
array registers = ["MAR","MDR","PC","ACC"]
print(registers[...])
```

write

Question 1

A linear search seeks 7 in [3,5,7,9]. Give number of checks. [2 marks]

Question 2

State a requirement of binary search. [1 mark]

Question 3

Explain what one pass of bubble sort guarantees. [3 marks]

Exit Ticket · Answer before checking

Linear search can work on an unsorted list.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

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Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. They show errors and often their location, helping the programmer find the cause.

Q2. A — Integer

Q3. A character represents one character; a string represents a sequence of characters.

Q4. Multiplication comes before addition, giving 10 in the original example. Operator precedence affects the result. Use brackets to make the intended order clear.

I Do / We Do pages · Guided practice

Q1. Index 3 found with one comparison.

You Do pages · Code First

```
array registers = ["MAR","MDR","PC","ACC"]
print(registers[2])
```

Expected result: PC

You Do pages · Independent questions

Q1. 3.

Q2. Sorted input.

Q3. After a full ascending pass, the largest unsorted value has reached the right end of that section.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed: